21. Port Output Enable for GPT (POEG)

21.1 Overview

The Port Output Enable (POEG) function can place the General PWM Timer (GPT) output pins in the output disable state in one of the following ways:

- Input level detection of the GTETRGn (n = A to D) pins
- Output-disable request from the GPT
- Comparator interrupt request detection
- Overcurrent Detection Window Notification detection (from DSMIF)
- Oscillation stop detection of the clock generation circuit
- Register settings

The GTETRGn (n = A to D) pins can be used as GPT external trigger input pins.

Table 21.1 lists the POEG specifications, Figure 21.1 shows a block diagram, and Table 21.2 lists the input pins.

Table 21.1 POEG specifications

Parameter	Specifications
Output-disable control through input level detection	The GPT output pins can be disabled when a GTETRGn rising edge or high level is sampled after polarity and filter selection.
Output-disable request from the GPT	When the GTIOCxA pin and the GTIOCxB pin are driven to an active level simultaneously, the GPT generates an output-disable request to the POEG. Through reception of these requests, the POEG can control whether the GTIOCxA and GTIOCxB pins are output-disabled.
Output-disable control through the comparator (ACMPHS) interrupt request detection	The GPT output pins can be disabled when an interrupt request is generated by a change in the output results of any of the comparators.
Output-disable control through overcurrent detection window notification detection	The GPT output pins can be disabled when overcurrent detection window notification from any channel of DSMIF is detected.
Output-disable control through oscillation stop detection	The GPT output pins can be disabled when oscillation of the clock generation circuit stops.
Output-disable control by software (registers)	The GPT output pins can be disabled by modifying the register settings.
Interrupt	Interrupts can be generated by detecting the input level of external trigger input pins (GTETRGn pins). Interrupts can be generated when all GPT or ACMPHS output pins are driven to an active level simultaneously. Interrupts can be generated when overcurrent detection window notification from any channel of DSMIF is detected.
External trigger output to the GPT	The GTETRGn signals can be output to the GPT after polarity and filter selection. (count start, count stop, count clear, up-count, down-count, or input capture function)
Noise filtering	For input from the GTETRGn pins, PCLKB/1, PCLKB/8, PCLKB/32, or PCLKB/128 can be selected as the noise filtering clock. (Filtering is performed by sampling the input signals three times using the selected clock.) Positive or negative polarity can be selected for any of the GTETRGn input pins. Signal state after polarity and filter selection can be monitored.
Module-stop function	Module-stop state can be set for each group to reduce power consumption.
TrustZone Filter	Security and Privilege attribution can be set for each group.

Note: n = A to D, x = 0 to 13

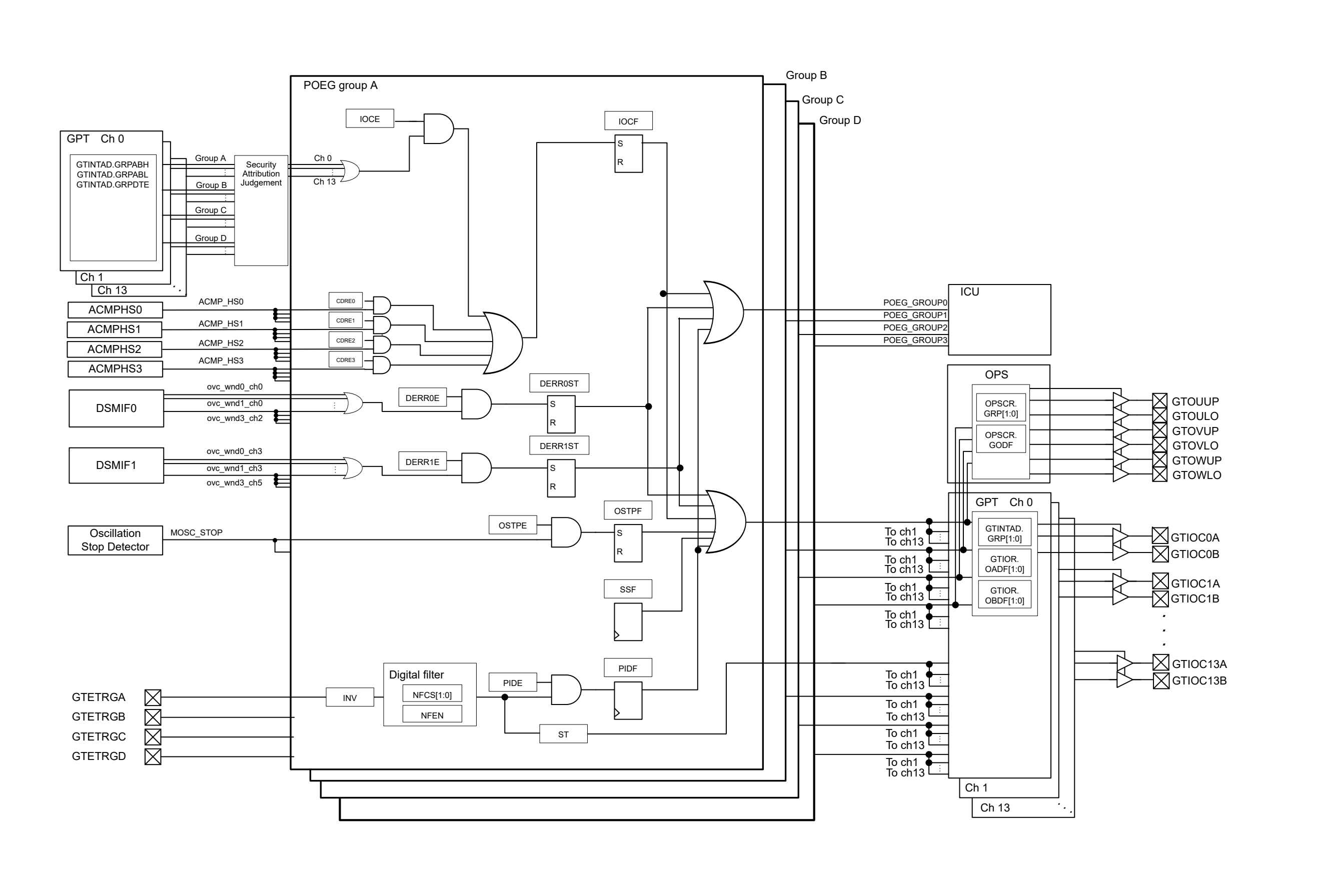


Figure 21.1 POEG block diagram

Table 21.2 POEG input pins

Pin name	I/O	Description
GTETRGA	Input	GPT output pin output-disable request signal or GPT external trigger input pin A
GTETRGB	Input	GPT output pin output-disable request signal or GPT external trigger input pin B
GTETRGC	Input	GPT output pin output-disable request signal or GPT external trigger input pin C
GTETRGD	Input	GPT output pin output-disable request signal or GPT external trigger input pin D

21.2 Register Descriptions

21.2.1 POEGGn : POEG Group n Setting Register (n = A to D)

Base address: POEG = 0x4021_2000
POEG_NS = 0x5021_2000

Offset address: 0x000 (POEGGA)
0x100 (POEGGB)
0x200 (POEGGC)
0x300 (POEGGD)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	NFCS[1:0]		NFEN	INV	DERR1E	DERR0E	DERR1ST	DERR0ST	—	—	—	—	—	—	—	ST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	CDRE3	CDRE2	CDRE1	CDRE0	—	OSTPE	IOCE	PIDE	SSF	OSTPF	IOCF	PIDF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PIDF	Port Input Detection Flag 0: No output-disable request from the GTETRGn pin occurred 1: Output-disable request from the GTETRGn pin occurred.	R/W*1
1	IOCF	Detection Flag for GPT or ACMPHS Output-Disable Request 0: No output-disable request from GPT or comparator interrupt occurred. 1: Output-disable request from GPT or comparator interrupt occurred.	R/W*1
2	OSTPF	Oscillation Stop Detection Flag 0: No output-disable request from oscillation stop detection occurred 1: Output-disable request from oscillation stop detection occurred	R/W*1
3	SSF	Software Stop Flag 0: No output-disable request from software occurred 1: Output-disable request from software occurred	R/W
4	PIDE	Port Input Detection Enable 0: Disable output-disable requests from the GTETRGn pins 1: Enable output-disable requests from the GTETRGn pins	R/W*2
5	IOCE	Enable for GPT Output-Disable Request 0: Disable output-disable requests from GPT 1: Enable output-disable requests from GPT	R/W*2
6	OSTPE	Oscillation Stop Detection Enable 0: Disable output-disable requests from oscillation stop detection 1: Enable output-disable requests from oscillation stop detection	R/W*2
7	—	This bit is read as 0. The write value should be 0.	R/W
8	CDRE0	ACMP_HS0 Enable 0: Disable output-disable request from ACMPHS0 1: Enable output-disable request from ACMPHS0	R/W*2
9	CDRE1	ACMP_HS1 Enable 0: Disable output-disable request from ACMPHS1 1: Enable output-disable request from ACMPHS1	R/W*2
10	CDRE2	ACMP_HS2 Enable 0: Disable output-disable request from ACMPHS2 1: Enable output-disable request from ACMPHS2	R/W*2
11	CDRE3	ACMP_HS3 Enable 0: Disable output-disable request from ACMPHS3 1: Enable output-disable request from ACMPHS3	R/W*2
15:12	—	These bits are read as 0. The write value should be 0.	R/W
16	ST	GTETRGn Input Status Flag 0: GTETRGn input after filtering was 0 1: GTETRGn input after filtering was 1	R
23:17	—	These bits are read as 0. The write value should be 0.	R/W
24	DERR0ST	DSMIF0 Error Detection Flag 0: No overcurrent detection window notification from DSMIF0 is detected 1: Overcurrent detection window notification from DSMIF0 is detected	R/W*1
25	DERR1ST	DSMIF1 Error Detection Flag 0: No overcurrent detection window notification from DSMIF1 is detected 1: Overcurrent detection window notification from DSMIF1 is detected	R/W*1
26	DERR0E	DSMIF0 Error Detection Enable 0: Disable output-disable request when overcurrent detection window notification from DSMIF0 is detected 1: Enable output-disable request when overcurrent detection window notification from DSMIF0 is detected	R/W*2
27	DERR1E	DSMIF1 Error Detection Enable 0: Disable output-disable request when overcurrent detection window notification from DSMIF1 is detected 1: Enable output-disable request when overcurrent detection window notification from DSMIF1 is detected	R/W*2

Bit	Symbol	Function	R/W
28	INV	GTETRGn Input Reverse 0: Input GTETRGn as-is 1: Input GTETRGn in reverse	R/W
29	NFEN	Noise Filter Enable 0: Disable noise filtering 1: Enable noise filtering	R/W
31:30	NFCS[1:0]	Noise Filter Clock Select 0 0: Sample GTETRGn pin input level three times every PCLKB 0 1: Sample GTETRGn pin input level three times every PCLKB/8 1 0: Sample GTETRGn pin input level three times every PCLKB/32 1 1: Sample GTETRGn pin input level three times every PCLKB/128	R/W

Note: S-TYPE3, P-TYPE3
Note 1. Only 0 can be written to clear the flag.
Note 2. Can be modified only once after a reset.

The POEGGn (n = A to D) registers control the output-disable state of the GPT pins, interrupts, and the external trigger input to the GPT.

In the descriptions, POEGGn represents the POEGGn (n = A to D) registers.

21.3 Output-Disable Control Operation

If any of the following conditions is satisfied, the GTIOCxA, GTIOCxB, and the 3-phase PWM output for BLDC motor control pins can be set to output-disable:

- Input level or edge detection of the GTETRGn pins
  When POEGGn.PIDE is 1, the POEGGn.PIDF flag is set to 1.
- Output-disable request from the GPT
  When POEGGn.IOCE is 1, the POEGGn.IOCF flag is set to 1 if the disable request is enabled by GTINTAD.
  The GTINTAD.GRPABH and GTINTAD.GRPABL settings apply to the group selected by the GPT register GTINTAD.GRP[1:0] or OPSCR.GRP[1:0].

  Note: The disable request is valid only when the security attributions of GPT and POEG are same. The disable request from GPT with a security attribution different from the security attribution of POEG is invalidated by the security attribution judgment.
- Comparator (ACMPHS) interrupt request detection
  Comparator interrupt detection is activated when any of the POEGGn.CDREi (i = 0 to 3) is 1. When the associated comparator interrupt is generated, the GPT output pins are disabled. POEGGn.IOCF indicates the detection status.
- Overcurrent Detection Window Notification detection (from DSMIF)
  While POEGGn.DERRiE (i = 0, 1) is 1, the overcurrent detection window notification is detected and POEGGn.DERRiST (i = 0, 1) is set to 1.
- Oscillation stop detection for the clock generation circuit
  While POEGGn.OSTPE is 1, the halt status of the main clock oscillator is detected and the POEGGn. OSTPF flag is set to 1.
- SSF bit setting
  When POEGGn.SSF is set to 1, the GPT and PWM output are disabled.

The output-disable state is controlled in the GPT module. The output-disable of the GTIOCxA and GTIOCxB pins is set in the GTINTAD.GRP[1:0], GTIOR.OADF[1:0], and GTIOR.OBDF[1:0] bits in GPTx. The output-disable of the 3-phase PWM output for BLDC motor control pins is set in the OPSCR.GRP[1:0] bits and OPSCR.GODF bit in GPT_OPS.

21.3.1 Pin Input Level Detection Operation

If the input conditions set in POEGGn.PIDE, POEGGn.NFCS[1:0], POEGGn.NFEN, and POEGGn.INV occur on the GTETRGn pins, the GPT output pins are output-disabled.

21.3.1.1 Digital Filter

Figure 21.2 shows high-level detection by the digital filter. When a high level associated with the POEGGn.INV polarity setting is detected three times consecutively with the sampling clock selected in POEGGn.NFCS[1:0], the detected level is recognized as high, and the GPT output pins are output-disabled. If even one low level is detected during this interval, the detected level is not recognized as high. In addition, in an interval where the sampling clock is not output, changes of the levels on the GTETRGn pins are ignored.

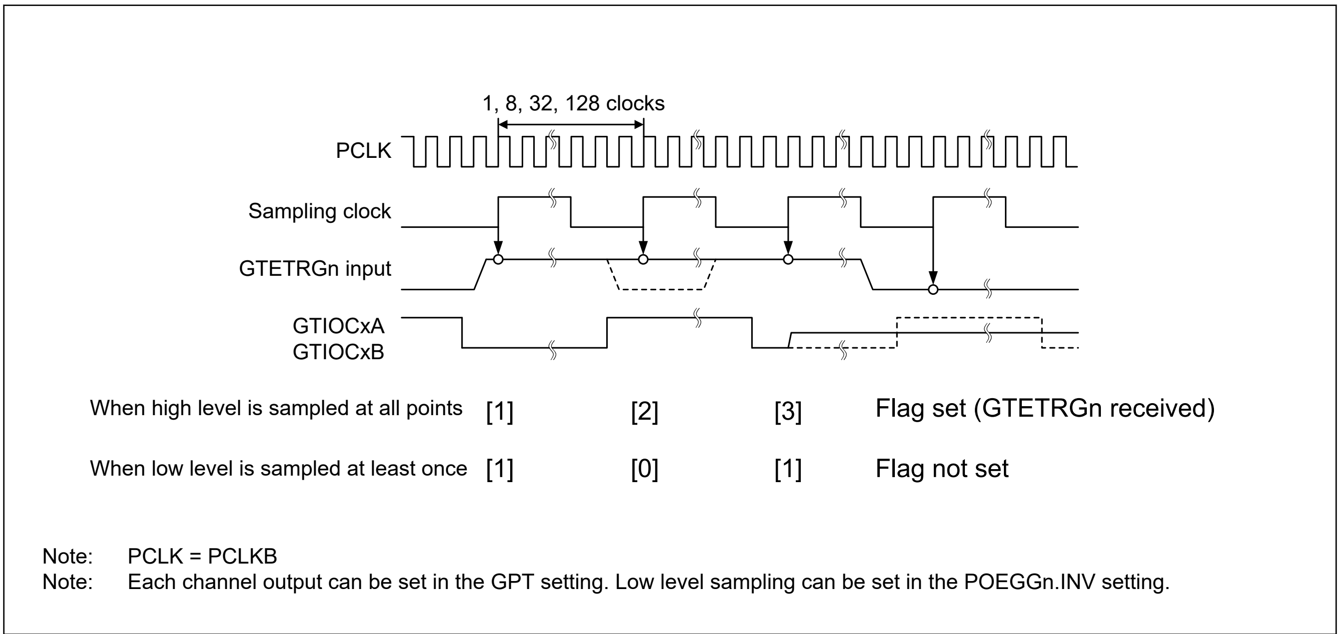


Figure 21.2 Example of digital filter operation

21.3.2 Output-Disable Requests from the GPT

For details on the operation, see the description for GTIOC Pin Output Negate Control in section 22, General PWM Timer (GPT).

21.3.3 Comparator Interrupt Detection

If POEGGn.CDREi (i = 0 to 3) is 1 when an associated comparator interrupt request is generated, the GPT output pins are output-disabled for each group. The status flag is POEGGn.IOCF which is shared with GPT output-disable detection.

21.3.4 Output-Disable Control Using Detection of Overcurrent Detection Window Notification

When overcurrent detection window notification occurred while POEGGn. DERRiE (i = 0, 1) is 1, the GPT output pins are output-disabled for each group.

21.3.5 Output-Disable Control Using Detection of Stopped Oscillation

When the oscillation stop detection function in the clock generation circuit detects stopped oscillation while POEGGn.OSTPE is 1, the GPT output pins are output-disabled for each group.

21.3.6 Output-Disable Control Using Registers

The GPT output pins can be directly controlled by writing 1 to the Software Stop flag, POEGGn.SSF.

21.3.7 Release from Output-Disable

To release the GPT output pins placed in the output-disable state, either return them to their initial state with a reset or clear all of the following flags:

- POEGGn.PIDF

- POEGGn.IOCF
- POEGGn.DERRiST (i = 0, 1)
- POEGGn.OSTPF
- POEGGn.SSF

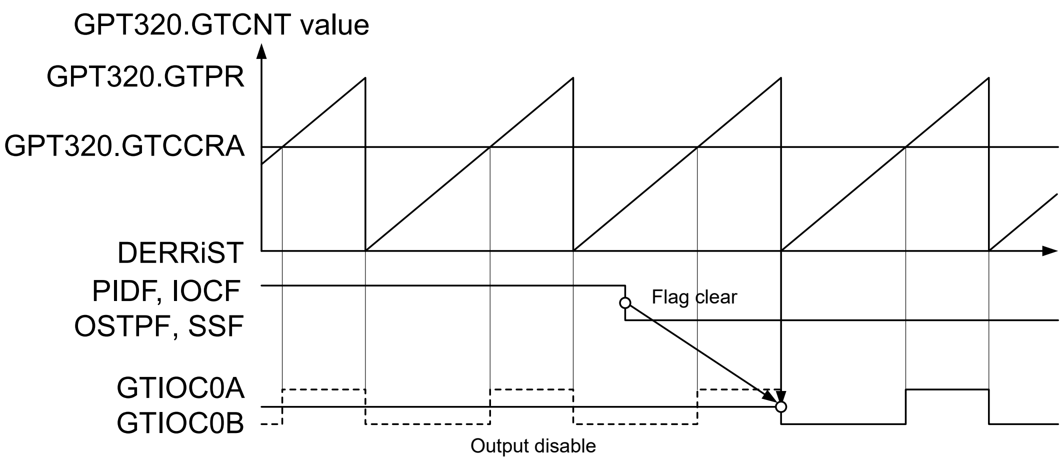
Writing 0 to the POEGGn.PIDF flag is ignored (the flag is not cleared) if the external input pins, GTETRGn are not disabled and the POEGGn.ST bit is not set to 0.

Writing 0 to the POEGGn.IOCF flag is valid (the flag is cleared) only if all of the GTST.OABHF and GTST.OABLF flags in the GPT are set to 0.

Writing 0 to the POEGGn.DERRiST (i = 0, 1) flag is valid (the flag is cleared) only if all of the DSCSRCHm.OWDxN (m = 0 to 2, x = 0 to 3) status bits in DSMIFi (i = 0, 1) is cleared.

Writing 0 to the POEGGn.OSTPF flag is ignored (the flag is not cleared) if the OSTDSR.OSTDF flag in the clock generation circuit is not set to 0. In addition, when the flag set and release occur at the same time, the flag set takes precedence.

Figure 21.3 shows the release timing for output-disable. The output-disable is released at the beginning of the next count cycle of the GPT after the flag is cleared.



Note: The base address of the registers are the same.

Figure 21.3 Output-disable release timing for GPT pin outputs

21.4 Interrupt Sources

The POEG generates an interrupt request for the following factors:

- Output-disable control by the input level detection
- Output-disable request from the GPT or the comparator (ACMPHS) interrupt request detection
- Output-disable request from DSMIF when overcurrent detection window notification detection

Table 21.3 lists the conditions for interrupt requests.

Table 21.3 Interrupt sources and conditions

Interrupt source	Symbol	Associated flag	Trigger conditions
POEG group A interrupt	POEG_GROUPA	POEGGA.IOCF	An output-disable request from a GPT disable request occurred
			An output-disable request from ACMPHS interrupt occurred
		POEGGA.DERRiST (i = 0, 1)	An output-disable request from DSMIF overcurrent detection window notification occurred
		POEGGA.PIDF	An output-disable request from the GTETRGA pin occurred
POEG group B interrupt	POEG_GROUPB	POEGGB.IOCF	An output-disable request from a GPT disable request occurred
			An output-disable request from ACMPHS interrupt occurred
		POEGGB.DERRiST (i = 0, 1)	An output-disable request from DSMIF overcurrent detection window notification occurred
		POEGGB.PIDF	An output-disable request from the GTETRGB pin occurred
POEG group C interrupt	POEG_GROUPC	POEGGC.IOCF	An output-disable request from a GPT disable request occurred
			An output-disable request from ACMPHS interrupt occurred
		POEGGC.DERRiST (i = 0, 1)	An output-disable request from DSMIF overcurrent detection window notification occurred
		POEGGC.PIDF	An output-disable request from the GTETRGC pin occurred
POEG group D interrupt	POEG_GROUPD	POEGGD.IOCF	An output-disable request from a GPT disable request occurred
			An output-disable request from ACMPHS interrupt occurred
		POEGGD.DERRiST (i = 0, 1)	An output-disable request from DSMIF overcurrent detection window notification occurred
		POEGGD.PIDF	An output-disable request from the GTETRGD pin occurred

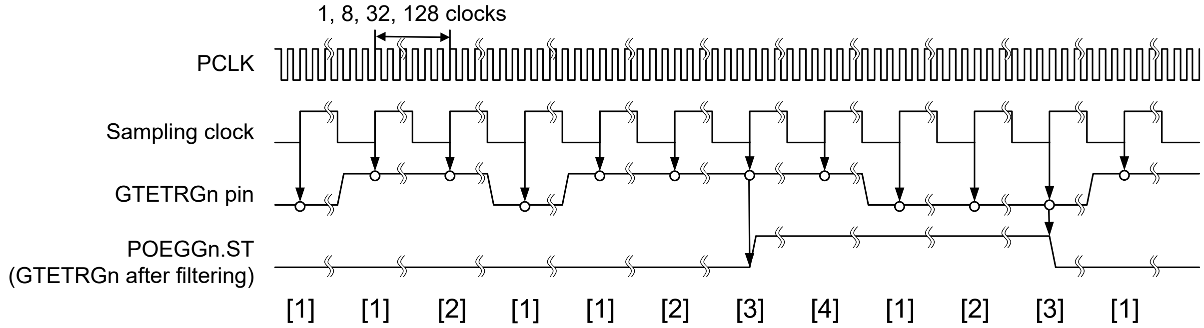
21.5 External Trigger Output to the GPT

The POEG outputs signals generated by filtering and level detection of GTETRGn pins input signals as the GPT operation trigger signal for the following:

- Count start
- Count stop
- Count clear
- Up-count
- Down-count
- Input capture

For the POEGGn.INV polarity setting signal, when the same level is input three times continuously with the sampling clock selected in POEGGn.NFCS[1:0], that value is output. Set the control registers the same as for the input level detection operation described in section 21.3.1. Pin Input Level Detection Operation The state after filtering can be monitored in POEGGn.ST.

Figure 21.4 shows the output timing of an external trigger to the GPT.



Note: PCLK = PCLKB
Note: Each channel output can be set in the GPT settings. Polarity can be reversed in POEGGn.INV.

Figure 21.4 Output timing of external trigger to the GPT

21.6 Usage Notes

21.6.1 Transition to Software Standby Mode

When using the POEG, do not invoke Software Standby mode. In this mode, the POEG stops and therefore output disable of the pins cannot be controlled.

21.6.2 Specifying Pins Associated with the GPT

The POEG controls output-disable only when a pin is associated with the GPT in the PmnPFS.PMR and PmnPFS.PSEL settings. When the pin is specified as a general I/O pin, the POEG does not perform output-disable control.